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Achieving competency in bronchoscopy: Challenges and opportunities

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ABSTRACT

Bronchoscopy education is undergoing significant changes in step with other medical and surgical specialties that seek to incorporate simulation-based training and objective measurement of procedural skills into training programmes. Low- and high-fidelity simulators are now available and allow learners to gain fundamental bronchoscopy skills in a zero-risk environment. Testing trainees on simulators is currently possible by using validated assessment tools for both essential bronchoscopy and endobronchial ultrasound skills, and more tools are under development for other bronchoscopic techniques. Educational concepts including the 'flipped classroom' model and problem-based learning exercises are increasingly used in bronchoscopy training programmes. These learner-centric teaching modalities require well-trained educators, which is possible thorough the expansion of existing faculty development programmes.

Key words: bronchoscopy, competency, education, interventional pulmonology, training.

Abbreviations: ACGME, American College of Graduate Medical Education; BSET, Bronchoscopy Stepwise Evaluation Tool; BSTAT, Bronchoscopy Skills and Tasks Assessment Tool; EBUS, endobronchial ultrasound; ICC, intraclass correlation; MCQ, multiple-choice question; PBL, problem-based learning; PA, Practical Approach; TBNA, transbronchial needle aspiration.

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INTRODUCTION

The rapidly evolving field of pulmonary procedures has led to unprecedented challenges in bronchoscopy education. With rising concerns over patient safety and cost-effective patient-centred care, the classical apprenticeship model by which patients suffer the burden of procedural training is currently ethically and legally unjustifiable. Opportunities for genuinely learner-centric training may be challenging due to limited exposure to novel technologies, work hour restrictions for trainees, lack of simulation equipment or facilities, and potential scarcity of skilled educators. Complications of basic bronchoscopy procedures are rare, but occasionally life threatening. Many bronchoscopy educators believe that physicians should strive for zero complications. To achieve this desirable goal, bronchoscopy training should meet certain standards to assure safety and effectiveness, which may be possible by implementing uniform training curricula. There is an ongoing paradigm shift in bronchoscopy education, and learning is becoming both more student- and patient-centric. Recent advances in the field of education research have highlighted the need for standardization of training and evaluation of trainees based on measurable competency metrics rather than mere procedural volume or subjective assessment from mentors and supervisors. Novel education tools such as simulation, flipped classroom models and problem-based learning (PBL) exercises are transforming the traditional approach to knowledge delivery. Successful implementation of these strategies requires investment in faculty development and supporting educators. In this article, we summarize the published evidence on simulation and validated assessment tools used in bronchoscopy, describe novel bronchoscopy teaching methodologies and present the objectives of available faculty development programmes pertinent to bronchoscopy education.

COMPETENCY-ORIENTED EDUCATION

Structured approach to bronchoscopy training

Bronchoscopy education should not be minimized to just procedural-based training as it encompasses heterogeneous and multifaceted components including the following: knowledge of thoracic anatomy, recognition of the scope of bronchoscopic interventions in the diagnosis and treatment of various respiratory pathologies, understanding of the technical aspects of specific bronchoscopy procedures, acquisition of the practical skillset required to manoeuvre the bronchoscope in the airway, perform biopsies or other sampling methods, and manage the complications of these procedures. Traditionally, trainees would start their education journey by presenting themselves to the bronchoscopy suite to be taught in a purely experiential manner, often with little theoretical foundations and almost always without any prior practical experience with the bronchoscope. Aside from the clear ethical dilemma raised by such an approach, this complete and sudden immersion experience in direct patient care would not uncommonly become overwhelming and a source of significant frustration and anxiety for both trainers and trainees. As such, novel approaches to bronchoscopy training have been developed. Structured learning curricula on websites, manuals and courses are now available providing the necessary theoretical and practical content in an accessible way.[1,2] Bronchoscopy learners can now be evaluated using standardized metrics for cognitive knowledge. A practical skillset is developed using bronchoscopy simulation, with measurable competency-based assessment tools ensuring adequate cognitive and procedural knowledge acquisition. Case-based scenarios as examples of PBL exercises can be incorporated on high-fidelity simulators and appropriate responses taught in a zero-risk environment. These components of training can be achieved prior to learners presenting for that first day in the bronchoscopy room.

The advantages of structured training programmes were highlighted in a review article published in 2000, which revealed that pulmonary trainees were only exposed to half the scope of procedures they might expect to perform in their clinical careers.[3] Since that time, there has been an exponential increase in the range of available procedures and hence an even greater need for structured training programmes.[4] Bronchoscopy and related procedures are generally safe, but complications, while relatively rare, can potentially be life threatening.[5,6] A recent study suggested an association between performance of procedures by trainees and incidence of complications and prolonged procedure times.[7] Studies in other specialties have demonstrated that simulation-based training before gaining experience in the clinical arena can mitigate these problems.[8–10] Structuring a training programme to incorporate optimal teaching methods will require wide dissemination of a core curriculum of theoretical knowledge, access to simulation training, ongoing validation of instruments to assess skills on simulators and then patients, access to training courses, and ultimately, confirming that clinical outcomes physician performance are indeed improved by such educational efforts.

Training programmes in other specialties use simulation routinely. These include, but are not limited to, anaesthesiology, surgery, gastroenterology and otorhinolaryngology.[11–16] The proven advantages of simulation are substantial and include reduced risks and discomfort for patients, increased exposure time of trainees to procedures and control of the complexity of training.[17] High-fidelity simulators often provide measurable competency metrics, some of which have been shown to discriminate between various levels of procedural expertise. A recent systematic review on simulation in bronchoscopy demonstrated clear benefits of simulation compared with no intervention in the areas of skills and behaviours, time and process outcomes.[18] There is evidence that evaluation and immediate feedback are a crucial component of effective learning and are in fact preferred by the majority of learners.[19–21] Such feedback is viewed as an opportunity to validate one's accomplishments while highlighting opportunities for improvement.[22]

To achieve these goals of competency-oriented metrics and objective immediate feedback, assessment tools for basic bronchoscopy and endobronchial ultrasound (EBUS) have been developed and validated.[23,24] In addition, assessment tools for other interventional pulmonary procedures including thoracic ultrasound and medical thoracoscopy have recently become available.[25–27]

One, 100 or 10 000 procedures?

Education scientists believe that the traditional model of 'see one, do one, teach one' is outdated, not learner or patient centric. Training guidelines from the American and European respiratory societies suggest procedural volume as an effective way of ascertaining adequacy of training.[28] For example, 100 flexible bronchoscopies and 50 EBUS procedures are currently proposed as adequate numbers to allow trainees to graduate from fellowship. Procedural volume was chosen to assess performance in these guidelines primarily because of the absence of more suitable alternatives. In fact, the evidence suggests that these arbitrarily defined numbers are inadequate as sole markers of effective training.[29,30] Trainees gain skills at different rates.[31] For example, in one study, bronchoscopy skill acquisition at specified milestones for 47 trainees showed a rapid upward trend in the first 20 procedures with a flattening of the learning curve thereafter, but there was considerable variation in skills acquisition at the 50th bronchoscopy.[22] The trainees assigned to the no-simulation group required many more procedures to achieve a level of competence similar to the simulation group. While these data would support establishment of a minimum number of procedures as a starting point of bronchoscopy training, they also highlight the need for competency-based metrics to account for the inter-individual variability in skill acquisition.[32] More recently, results from a multicentre study suggest that pulmonary trainees need an average of 13 procedures to achieve their first independent successful performance of EBUS-transbronchial needle aspiration (TBNA).[33] However, because trainees learn at different pace, have different depth perception skills, manual dexterity and ability to adapt to new situations, the use of absolute numbers to declare a physician competent in a specific procedure makes little sense. It has become evident that while a minimum number of procedures may still be required for accreditation and credentialing, the future of procedural competency research will focus on the development of validated assessment tools as objective measures of performance.[32] It is probably true that the more procedures one performs, the better one becomes. It is known from other domains that success and achievement are only partially due to innate talent and opportunity but are mostly determined by the degree of preparation and practice. This is true in computer science, sports, literature and music. In fact, it has been suggested that approximately 10 000 h of practice are warranted to become an elite performer in a complex task.[i,34] This level of mastery, however, may be limited to a few world-class experts in every field. In a medical procedure-oriented field like interventional pulmonology, short of waiting until one is at least mid-career, the '10 000 h rule' cannot constitute a feasible requirement to certify physicians in specific procedures. On the other hand, choosing arbitrary numbers as it has been done so far is not a fair alternative either. There is therefore an urgent need for validated assessment tools and checklists that are procedure specific or even task specific and are designed to measure performance metrics.

Assessment tools and checklists

Competency-based assessments are a prolific area of education research and are increasingly adopted by many training programmes.[35] Validated assessment tools represent the cornerstone of competency-based programmes and allow continuous feedback highlighting improvements and remaining opportunities for improvement.[36] It is also fair to expect that regulatory agencies, third-party payers and patient interest groups will continue to demand increasingly rigorous guarantees that medical trainees are adequately trained and their level of competency objectively assessed before being allowed to independently engage in patient care.[37]

Investigators have developed the concept of competency-based metrics in bronchoscopy training.[29,38,39] This multistep process consists of the following: competency-based metrics development, assessment tool validation of accuracy and reproducibility, and implementation of these validated tools in the clinical area. These studies have in part used the materials available on the Bronchoscopy International website (http://www.bronchoscopy.org) as the basis for theoretical and practical learning.[1,40,41] This website contains information on a wide range of bronchoscopy topics and includes instructional videos, assessment tools as well as self-directed sets of multiple-choice questions (MCQ). The development of a free-to-use and widely available educational resource such as the Bronchoscopy International website is particularly critical in the developing world and ensures democratization of knowledge.[42]

In 2008, two instruments were studied and validated for the assessment of bronchoscopy skills: the Bronchoscopy Skills and Tasks Assessment Tool (BSTAT) and the Bronchoscopy Stepwise Evaluation Tool (BSET).[43] BSTAT assesses basic bronchoscopic anatomy knowledge, facility in manoeuvring the bronchoscope through the bronchial tree and performing tasks such as biopsy or bronchoalveolar lavage, and the identification of common airway pathologies. The BSET tests learner's knowledge of a series of specific manoeuvres designed to facilitate bronchoscope handling and comprises a global rating scale and a rating of the proper execution of the specific manoeuvres. In one study, two examiners scored 22 subjects using a high-fidelity simulator. The intraclass correlation (ICC, interrater reliability) was tested on results within the different groups (novice, intermediate and advanced), and test–retest reliability was measured at repeat testing on each candidate after at least 2 weeks. There was a very high level of ICC (0.98) for both the BSAT and BSET. There was also good test–retest reliability with ICC sores of 0.86 and 0.85 for BSTAT and BSET, respectively. There were clear differences in absolute mean scores for novices versus experienced bronchoscopist, and although results for fellows at an intermediate level overlapped considerably with those of consultants, there was a non-significant trend towards better scores in experienced operators.

Competency-based assessment tools may be useful to assess the efficacy of training courses. Colt and colleagues applied their MCQ test and the BSTAT before and after a 1-day training course with 24 enrolled learners.[39] Pretest and post-test results were significantly improved with an absolute gain of 18% for MCQ and 34% for BSTAT. The class average normalized gain (the average actual gain divided by the maximum possible gain) was 34% for MCQ and 60% for BSTAT. A predefined target of 30% was set as the minimum value for the intervention to be regarded as effective.[44] An interesting point in this study was that the course was rated very highly by attendees in spite of substantial testing time, often considered a deterrent or at best an unnecessary distraction during training courses. This suggests that when adequately administered, low-stakes evaluations may be viewed as valuable components of a training programme by both trainers and trainees.[45,46]

[i]This refers to famous hockey players, musicians and computer scientists; no data are available to justify such numbers in the medical field.

A multicentre prospective study of 47 fellows used the same evaluation tools to assess practical and cognitive bronchoscopy skills in training at predefined milestones up to 100 patient bronchoscopies.[22] In the simulation training group, 20 simulation bronchoscopies had to be performed by trainees before the fifth real patient bronchoscopy. Simulation training resulted in a statistically significant improvement in BSTAT scores in seven of the eight milestone bronchoscopies. This improvement was particularly evident in the first 30 bronchoscopies and supports the integration of such tools for bronchoscopy training in pulmonary fellowship programmes. By contrast, there was considerable variation in skill acquisition level for those who had 'conventional' training.

Advanced diagnostic bronchoscopy procedures have been the object of fewer reports. A competency-based assessment tool for EBUS was developed and validated.[47] This is a 10-section assessment tool incorporating anatomy, equipment handling, and computed tomography and EBUS image interpretation. In an initial study, 24 bronchoscopists were divided into three groups based on the number of prior EBUS-TBNA cases done before testing on patient procedures. The test discriminated well among beginners, intermediate and experienced proceduralists with statistical significance. There was a linear relationship in test scores for those with a prior experience of less than 50 procedures. There was no benefit to a candidate having done more than 50–100 procedures in terms of EBUS Skills and Tasks Assessment Tool (STAT) scores because the test was designed to have a ceiling of scores.[48–50] As such, it is likely that further improvement towards procedural excellence would not be suitably assessed by the EBUS STAT[51] Wahidi *et al.* have recently studied the time course of EBUS-TBNA skill acquisition in trainees over a 2-year period.[33] The majority of trainees performed all the essential steps in EBUS-TBNA and obtained adequate tissue after an average 13 procedures (range 7–16).

Other competency-based metrics have been proposed and others are being developed for a variety of bronchoscopy and interventional pulmonology procedures. In a series of studies by Konge and colleagues, a pass/fail approach to bronchoscopy simulation testing was developed.[52–54] The authors used an instrument that included testing of manual dexterity, anatomy and endobronchial image recognition components. The test was applied to experts (more than 500 bronchoscopies) and 14 medical students. A pass score was determined as the point at which the score distributions of the two groups met. In subsequent testing, 17 trainees were tested, and all but one trainee who had done more than 75 bronchoscopies passed the test.[54]

We believe that competency-based assessment tools are relevant instruments for the objective and standardized evaluation of bronchoscopy trainees. Validated tools are available and easily accessible. They directly address the limitations of a classical procedural volume-based method of evaluation. As such, in our opinion, they should become part of a uniform approach to bronchoscopy training and integrated into future iterations of training guidelines by professional societies.

LEARNER-CENTRIC TEACHING METHODS

The traditional classroom lecture-based method of teaching has been called into question by many authors, and alternative teaching methods are gaining acceptance.[55,56] New student-centred teaching methodologies are being implemented in medical schools. Constructive feedback is a critical part of the new training focus, and simulation and testing provide a platform for this to occur. In one study, for instance, a group of trainees was tested on a high-fidelity simulator and was randomized to receiving constructive feedback by an experienced supervisor or just complete the tasks. A range of metrics was significantly better in the feedback group.[57] In this section, we briefly discuss two teaching modalities that encourage learner-teaching contact time to address learner's needs: the 'flipped classroom' concept and PBL exercises, which have already been applied to bronchoscopy-oriented training.

Flipped classroom model

Shortcomings of the current medical education model are multiple and include: learning being focused on test performance rather than developing professional competencies; grading methods not reflecting true physicians' skills, attributes or behaviours; and assessments of students by faculty being subjective and not reflecting critical thinking skills.[58] A paradigm shift is offered by what has been called 'disruptive innovations' such as the 'flipped classroom' and massive online open courses. While in one form or another the 'flipped classroom' has existed for decades, it has only recently become popular when, with the help of web-based technology, it was introduced for high school students who missed their classroom lessons. This instructional design has since gained significant popularity in many adult learning environments. In this teaching model, traditional lectures can be provided to students online for reading in their own time, leaving actual student–teacher contact time for engaging in PBL and enquiry. In the medical field, for example, one study 'flipped' a pharmacy course, delivering the lectures online and allowing the interactive activities in 'class' time.[55] The strategy was very much enjoyed by students, and overall class attendance improved. Enjoying the class, however, may be irrelevant, as long as this teaching modality proves its effectiveness. In this regard, a study of first year physiology in cardiovascular respiratory and renal disease students showed a 12% higher multiple choice test scores in a 'flipped classroom' setting compared with traditional lectures. This teaching methodology has been successfully implemented at international bronchoscopy courses organized by the American College of Chest Physicians.[59] This teaching philosophy was not designed to be dogmatic. The core concept is to flip the traditional instructional approach. Content materials such as videos, PowerPoint presentations and relevant articles are accessed at home prior to class. On their own, these precourse materials may not be valuable, but when integrated appropriately they become valuable to the learner. For instance, the teachers should ask the students to take notes after watching these materials and come to the class with questions. The classroom becomes a place to work through problems and engage in collaborative learning, which can be done through PBL exercises. In the class, teachers and students connect through hands-on activities, projects and tutoring. In this model, learners take responsibility for their learning both inside and outside the classroom. This is similar to humanities professors' teaching who expect their students to read a novel on their own and use the class time to explore symbols and analyse themes. The emphasis is not on transmission of information, but on helping students assimilate and apply that information.[60] The potential resistance to this concept in the medical procedural education may be determined by at least two factors: (i) existing courses have to be redesigned and (ii) new curricula have to be taught by quality teachers. That is why recruiting, training and supporting quality educators for this type of instructional design becomes relevant. The concept of flipping the classroom is promising and timely as policy-makers and advocacy groups who seek to improve education want concrete evidence that students are truly learning.

PBL exercises

PBL is an example of constructivism[ii]-based learning environment. In some views, it is characterized by three fundamental propositions: (i) understanding is in our interactions with the environment; (ii) cognitive conflict or 'puzzlement' is the stimulus for learning; and (iii) knowledge evolves through social negotiation and through the evaluation of the viability of individual understandings.[61] In constructivism, understanding is a function of the content, the context, the activity of the learner and the goals of the learner, all of which are addressed in PBL exercises. The stimulus for learning is a primary factor in determining what the learner attends to, what prior experience he or she brings in constructing an understanding. Collaborative groups (in the form of team exercises) are important for learning because one can test his or her own understanding and examine the understanding of others as a mechanism for enriching and expanding the understanding of particular issues. These propositions suggest a set of instructional principles applied in designing PBL environments.

PBL was developed in medical education in the early 1970s, and since that time it has been refined and implemented in medical, business, law, architecture and engineering education. In medicine, these exercises help learners gain cognitive, technical and experiential skills in a patient- and learner-centric environment. One such model, the 'Practical Approach (PA) to Interventional Bronchoscopy', is based on a four-box format inspired from Albert Jonsen' s work on clinical ethics.[62] This type of PBL has been developed to complement traditional learning models as it helps learners think about the reasons of their actions and implementation strategies based on background information, relevant literature and personal experience. As with any instructional model, there are many strategies for implementing PBL. This teaching modality has been used in lectures, articles and textbooks in order to help learners deconstruct a procedure-related consultation into its basic elements.[63] It is used as a dynamic teaching tool in several Pulmonary and Critical Care Medicine training programmes. In addition, a series of PowerPoint presentations addressing a variety of problems in interventional bronchoscopy use the PA format and are available online free of charge.[64] This methodology can be implemented to any procedure-oriented training. In fact, the PA format follows the current recommendations from the American College of Graduate Medical Education (ACGME) for training programmes through which trainees should learn to gather essential and accurate information about their patients, make informed decisions about diagnostic and therapeutic interventions based on patient information, preferences, up-to-date scientific evidence and clinical judgment, use information technology to support patient care decisions and patient education, develop patient management plans, counsel and educate patients and their families, communicate effectively and demonstrate caring and respectful behaviours when interacting with patients and their families, and work with health-care professionals, including those from other disciplines, to provide patient-focused care .[65]

The three major elements of a procedure are addressed in the PA format type of PBL: strategy and planning, execution and response to procedure-related adverse events or complications. Using the four-box approach (Table 1), the learner may address in the desired detail the major elements of the case at hand. Any subject matter pertinent to procedural training can be addressed using this model, depending on the learners' needs. Each PA exercise begins with a real case description with the images and clinical scenarios modified to assure patient's privacy and to provide an authentic learning environment.[iii] Every case element can be addressed: physical examination and complementary tests; procedure indications, risks and alternatives; anatomy, anaesthesia and perioperative care; results, complications and outcome assessments; patients preferences, support systems and expectations; ethics and palliative care; and team experience and quality improvement. The depth of analysis of these items varies from case to case, depending on the teaching points that need to be emphasized. For an example on how to use this

[ii]Constructivism is a philosophical view on how we come to understand or know.

[iii]An authentic learning environment is one in which the thinking required is consistent with the cognitive demands in the environment for which the learner is preparing for.

Table 1 The four-box approach for problem-based learning

Initial evaluation	Procedural strategies
1. Physical examination, complementary tests and functional status assessment 2. Patient's significant comorbidities 3. Patient's support system (also includes family) 4. Patient preferences and expectations (also includes family)	1. Indications, contraindications and expected results 2. Operator and team experience and expertise 3. Risk-benefits analysis and therapeutic alternatives 4. Respect for persons (informed consent)
Procedural techniques and results	**Long term management plan**
1. Anaesthesia and other perioperative care 2. Techniques and instrumentation 3. Anatomic dangers and other risks 4. Results and procedure-related complications	1. Outcome assessment 2. Follow-up tests, visits and procedures 3. Referrals to medical, surgical or palliative/end of life subspecialty care 4. Quality improvement and team evaluation of clinical encounter

teaching modality, the readers are referred to a free online instructional video, which can also offer a framework for using the four-box model to review cases in daily practice as well as a method for self-evaluation.[66]

PBL exercises can be completed alone or as a group with or without the guidance of an instructor. To better address the ACGME's requirements to teach learners how to work with other health-care professionals, including those from different disciplines, and to provide patient-focused care, this model has been implemented at national and international courses and faculty development programmes. The aim is to encourage teamwork in diagnosis and management of a certain clinical problem.[67,68] For the purpose of an educational programme using a 'flipped classroom' model, the cognitive skills are enhanced by reading or by interactive sessions, and technical skills by using simulation. The main advantage of the PBL is that these two forms of knowledge can be combined with experiential learning. Working through a case-based exercise helps learners contemplate a set of patient- or procedure-related issues in a structured fashion. The purpose of the learning activity, however, must be clear to the learner so that he or she can see it as relevant and be engaged in the project. Therefore, the PBL exercise has to be designed and facilitated in a way that the learners will adopt the problem as their own. While alone or in a group, the learners work through the clinical scenario as they might work through an academic pulmonary consultation. In a group format addressing subject matters such as lung cancer, the PBL offers a tumour board-like environment and has already been implemented on several venues.[59] The role of the instructor facilitating the PBL exercise is to engage the learners in meaningful dialogue to help bring the task home to the learner.[61] Once the case resolution is completed, a team leader presents the case to the other participants and feedback is obtained. To answer the question 'How would the learners know that this exercise worked for them?' assessment tools have been developed and implemented during their work as a team and after the team leader's presentation.

PBL exercises aim to materialize the concept of *adaptive expertise*[iv] in which a new context becomes relevant to learning. Specific principles and concepts are transferred through activities and not through formal didactic lectures. These exercises allow the trainers to test the learner's ability to adapt to novel situations by applying previously gained knowledge for new tasks. Thus, the learners can be assessed on their knowledge flexibility. By offering learners the opportunity to break from routines, innovate and explore different pathways, *adaptive expertise* exercises in the form of PBL become interesting learning experiences. PBL exercises encourage collaboration, which has been suggested to help develop *adaptive expertise*. As learners work together in a group, each member expresses their individual ideas. As group members listen to each other's ideas, they are encouraged to reflect on and reconsider their viewpoints.[69] Learning scientists believe that effective learning environments should emphasize a learning community where ideas are discussed and understanding is enriched. PBL offer such a venue. We believe that PBL in the PA four-box format offers the complexity of the practice environment that the learners are expected to function in when they return to their clinical practice. They become engaged and gain ownership of the process.[61] The instructors are merely facilitators who challenge the learners' thinking without dictating it. The facilitators are also designated to be available as consultants, as they would be for any physician in the real world. Thus, the faculty assumes a major role in modelling the learner's thinking associated with the problem-solving process. Faculty development programmes are therefore essential for assuring the success of PBL implementation. Studies are war-

[iv]Adaptive expertise is a broad construct that encompasses a range of cognitive, motivational and personality-related components, as well as habits of mind and dispositions. Problem solvers demonstrate adaptive expertise when they are able to efficiently solve previously encountered tasks and generate new procedures for new tasks (http://en.wikipedia.org/wiki/Adaptive_expertise; accessed 1 Feb 2014).

ranted, however, to assess the benefits of this teaching model for adult learning relevant to procedure-oriented training.

BRONCHOSCOPY SIMULATION: HYPE OR HOPE?

While offering distinct advantages, such as a one-on-one interaction between trainer and trainee and potentially highly personalized education that may fit better with a particular trainee's unique learning pace and preferences, the Halstedian education model of bronchoscopy training is now considered inadequate for several reasons. First, heightened concerns over patient safety, highlighted by a landmark report published in 1999 by the Institute of Medicine, have led to increased scrutiny of medical training in the public opinion.[70] Furthermore, work hour restriction rules in medical training programmes are radically reshaping the way knowledge is delivered to trainees, limiting their exposure to routinely and, *a fortiori*, less commonly performed procedures, and mandating more standardized methods for training and evaluation of learners. Finally, complications of basic bronchoscopy, some of which are life threatening, are rare and unlikely to be encountered during conventional clinical training. Yet they require the skillset needed to respond to these situations appropriately. Because simulation effectively deals with all these challenges, it is quickly becoming a cornerstone of bronchoscopy training.

Searching high and low for the optimal simulation model

Published evidence on simulation training in bronchoscopy remains relatively modest, but the need for product development has clearly become evident in the recent years, and multiple systems are now available on the marketplace. Bronchoscopy simulators are generally described as either high or low fidelity. While these descriptors refer to the degree of accuracy and realism of the simulation experience, they are somewhat of a misnomer in that low-cost, 'low-fidelity' interfaces may on occasions be viewed by trainers and trainees more convincing than their high-fidelity counterparts.[38,71] High-fidelity systems consist of a proxy bronchoscope introduced in an interface with sensors connected to a computer (Fig. 1). The movements of the bronchoscope are analysed in real time, and three-dimensional virtual reality images are displayed on a computer screen with haptic and auditory feedback.[20] A distinct advantage of these systems lies is the degree of realism of the experience, with a variety of clinical scenarios available and real-life responses from virtual patients who breathe, cough and may develop complications. In addition, high-fidelity simulation models provide detailed analyses of trainees' performance with objective metrics (such as number of segments entered, wall collisions, dose of lidocaine used and adequate sampling methods). Several high-fidelity

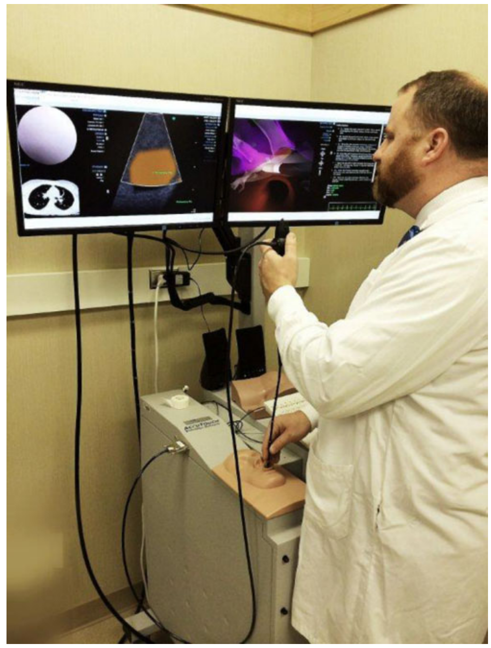

Figure 1 Example of practice on a high-fidelity simulator.

simulators are available, most with a cost exceeding \$100 000, and offer the entire spectrum of basic bronchoscopy procedures such as the endoscopy VR simulator (CAE Healthcare, Montreal, Quebec, Canada) and the GI-BRONCH Mentor (Simbionix Ltd, Airport City, Israel). EBUS modules are also available and facilitate acquisition of advanced cognitive and technical skills before proceeding to real patients.[72,73] More affordable high-fidelity simulators are also in development and should allow easier access to these technologies.

Low-fidelity simulation models are generally cheaper and more readily available, and consist of inanimate platforms with variable degrees of realism. They also allow trainees to use real bronchoscopes, which may augment the realism of the procedure. An interesting model is the Dexter model (Replicant, Wellington, New Zealand), with branching tubular structures similar to the bronchial tree, but without direct correlation to human anatomy.[9,74,75] The purpose of this system is to emphasize psychomotor skills acquisition without distraction from the lack of knowledge of the anatomy in the early phases of training. Other low-fidelity systems range from anatomically accurate synthetic models to non-anatomic inexpensive 'choose-the-hole' box.[9,13,22,38,71] There are relatively few studies assessing the impact of low-fidelity compared with high-fidelity simulation training on bronchoscopy skills, but a recent meta-analysis suggests that low-fidelity models may in fact be preferred in some situations, even though more interesting models may ultimately motivate learners to practice more.[18] Keeping the limited available evidence in mind, low-fidelity models may be more suitable for the most basic bronchoscopy tasks while simulation of more advanced procedures, such as EBUS and navigation bronchoscopy, are likely to be offered by high-fidelity systems in the future.

Simulation and bronchoscopy training: a summary of the evidence

A recently published systematic review and meta-analysis on simulation-based bronchoscopy training included 17 comparative studies and unsurprisingly concluded that in comparison with no intervention, substantial benefits were achieved with simulation in many learning outcomes including skills, behaviours and time.[18] The majority of published comparative studies have only looked at basic flexible bronchoscopy skills, specifically inspection bronchoscopy. It would intuitively seem appealing to recommend simulation training for more complex and less commonly performed procedures. Yet only few studies have evaluated simulation in rigid bronchoscopy, EBUS or conventional TBNA training.[13,38,72,73,76] Interestingly, one EBUS study is one of the few studies supporting transferability of simulation-acquired procedural skills in the clinical practice, supporting simulation training for advanced procedures.[72] Another important point to highlight is that none of these studies have assessed a mastery model of learning, by which evidence of mastery of one step is required before advancing to the next level. This model has been shown to be very effective, is already used in bronchoscopy education curriculum ('Bronchoscopy International, step-by-step') and should be considered in future bronchoscopy simulation studies.

Published studies have compared simulation-based bronchoscopy training with (i) no intervention, (ii) non-simulation training and (iii) alternative simulation strategies.[18] Compared with no training, simulation results in large benefits in process outcomes (procedural skills) with a pooled effect size of 1.21 (95% confidence interval: 0.82–1.60, $P < 0.01$). Two of these eight studies reported similar effect size for procedural outcomes in real patients, supporting the clinical transferability of the acquired skills.[22,77] Interestingly, simulation training may actually increase procedural time for trainees, but with improvement of the thoroughness and quality of the procedure.[77] Compared with non-simulation training (i.e. conventional clinical or purely didactic training), the effect of simulation on process and time were substantial, but the small sample size of the included studies did not allow for definitive conclusions. Of more importance to educators, a few implications of this systematic review are deducible from the few studies comparing different simulation strategies.[9,38,71,78] While the strength of the evidence is limited, the data available suggest that low-fidelity models (in studies addressing basic bronchoscopy skills) may actually be considered as more realistic by learners, but also that the more interesting the models are perceived, the more stimulated the learners are. Highly structured educational programmes, which include both simulation and formal didactic and case-based lectures, are preferred by learners.

FACULTY DEVELOPMENT PROGRAMMES

Faculty development initiatives may involve the training of senior fellows or established practitioners. One model for advancing the principles of simulated bronchoscopic training for fellows has been provided by the 'Senior Tour' concept in cardiothoracic surgery, under the auspices of the Thoracic Surgery Directors Association, and the Joint Council on Thoracic Surgery Education.[79] The Senior Tour was devised to generate a critical mass of senior surgical educators who are familiar with simulation in training. One role senior fellows can have is in suggesting reasonable levels of procedural efficacy based on outcomes.[8,21]

Bronchoscopy International has pioneered projects specifically designed to help bronchoscopists gain in-depth knowledge of a structured, uniform curriculum pertaining to introductory courses in flexible bronchoscopy. These 'Train the trainers' programmes have the goal to establish a minimum standard of bronchoscopic factual and procedural skills. These programmes provide learners with the opportunity to improve their teaching skills, practice teaching, work as a team in a learner-centric environment and be prepared to establish training programmes and workshops on their own.[80] Faculty development programmes for bronchoscopists are underway around the world under the auspices of the World Association Bronchology and Interventional Pulmonology and Bronchoscopy International.[67,68] Such programmes have been already implemented in several countries from North and South America, Asia and Europe. They use the 'flipped classroom' teaching philosophy, through which reading material such as the Faculty Development Program manual, Introduction to Flexible Bronchoscopy curriculum, selected articles on education philosophies, PowerPoint presentations, video exercises and the textbook *The Essential Flexible Bronchoscopist* can be studied at leisure prior to the course. During the actual seminar, the emphasis is on *hands-on* experience whereby future bronchoscopy educators learn to use low- and high-fidelity models for teaching fundamental bronchoscopy; demonstrate the use of validated assessment tools; learn how to use PBL exercises in a structured curriculum; demonstrate the use of MCQ in interactive didactic sessions; and work as a team to design a structured bronchoscopy training curriculum. The long-term educational impact of these programmes will be a matter of further research.

CONCLUSIONS AND FUTURE OPPORTUNITIES

The current published data and education theories support the integration of simulation, competency-based metrics, flipped classroom model and PBL in bronchoscopy training programmes. Potential barriers to implementation include cost issues, the question of clinical transferability of procedural skills acquired in simulation models, the absence of proven improvement in actual patient clinical outcomes and the general resistance to change ('we've always done it that way'). Low-fidelity models may be preferred to more expensive computer-based platforms in selected situations. Cheaper virtual reality models are under development, which should mitigate the impact of some of the cost barriers. Several bronchoscopy education resources are widely available at low or no cost. Novel education tools such as online courses, PBL and flipped classroom models constitute exciting developments in education and are very much needed in bronchoscopy education. There is a need for recruiting, training and supporting skilled faculty for implementing such learner-centric programmes. This is now possible through faculty development programmes offered by national and international chest medicine organizations. With further validation of curricula, teaching strategies, assessment tools and faculty development programmes, we envision the widespread acceptance of these concepts and uniform bronchoscopy education in thoracic medicine training programmes around the world.

REFERENCES

1 Bronchoscopy International. The essential bronchoscopist. 2013. [Accessed 2 Feb 2014.] Available from URL: http://www.bronchoscopy.org/
2 Mata CA, Ota LH, Suzuki I, Telles A, Miotto A, Leão LE. Web-based versus traditional lecture: are they equally effective as a flexible bronchoscopy teaching method? *Interact. Cardiovasc. Thorac. Surg.* 2012; **14**: 38–40.
3 Haponik EF, Russell GB, Beamis JF Jr, Britt EJ, Kvale P, Mathur P, Mehta A. Bronchoscopy training: current fellows' experiences and some concerns for the future. *Chest* 2000; **118**: 625–30.
4 Musani AI, Gasparini S. Advances and future directions in interventional pulmonology. *Clin. Chest Med.* 2013; **34**: 605–10.
5 Asano F, Aoe M, Ohsaki Y, Okada Y, Sasada S, Sato S, Suzuki E, Senba H, Fujino S, Ohmori K. Deaths and complications associated with respiratory endoscopy: a survey by the Japan Society for Respiratory Endoscopy in 2010. *Respirology* 2012; **17**: 478–85.
6 Eapen GA, Shah AM, Lei X, Jimenez CA, Morice RC, Yarmus L, Filner J, Ray C, Michaud G, Greenhill SR *et al.*, American College of Chest Physicians Quality Improvement Registry, Education. Complications, consequences, and practice patterns of endobronchial ultrasound-guided transbronchial needle aspiration: results of the AQuIRE registry. *Chest* 2013; **143**: 1044–53.
7 Stather DR, MacEachern P, Chee A, Dumoulin E, Tremblay A. Trainee impact on procedural complications: an analysis of 967 consecutive flexible bronchoscopy procedures in an interventional pulmonology. *Pract. Respir.* 2013; **85**: 422–8.
8 Training Committee 2010–2011, Sedlack RE, Shami VM, Adler DG, Coyle WJ, DeGregorio B, Dua KS, DiMaio CJ, Lee LS, McHenry L Jr *et al.* Colonoscopy core curriculum. *Gastrointest. Endosc.* 2012; **76**: 482–90.
9 Martin KM, Larsen PD, Segal R, Marsland CP. Effective nonanatomical endoscopy training produces clinical airway endoscopy proficiency. *Anesth. Analg.* 2004; **99**: 938–44.
10 Fried GM, Feldman LS. Objective assessment of technical performance. *World J. Surg.* 2008; **32**: 156–60.
11 Hatton KW, Price S, Craig L, Grider JS. Educating anesthesiology residents to perform percutaneous cricothyrotomy, retrograde intubation, and fiberoptic bronchoscopy using preserved cadavers. *Anesth. Analg.* 2006; **103**: 1205–8.
12 Ansell J, Mason J, Warren N, Donnelly P, Hawkes N, Dolwani S, Torkington J. Systematic review of validity testing in colonoscopy simulation. *Surg. Endosc.* 2012; **26**: 3040–52.
13 Jabbour N, Reihsen T, Sweet RM, Sidman JD. Psychomotor skills training in pediatric airway endoscopy simulation. *Otolaryngol. Head Neck Surg.* 2011; **145**: 43–50.
14 Sutton E, Chase SC, Klein R, Zhu Y, Godinez C, Youssef Y, Park A. Development of simulator guidelines for resident assessment in flexible endoscopy. *Am. Surg.* 2013; **79**: 14–22.
15 Ishman SL, Brown DJ, Boss EF, Skinner ML, Tunkel DE, Stavinoha R, Lin SY. Development and pilot testing of an operative competency assessment tool for pediatric direct laryngoscopy and rigid bronchoscopy. *Laryngoscope* 2010; **120**: 2294–300.
16 Malekzadeh S, Malloy KM, Chu EE, Tompkins J, Battista A, Deutsch ES. ORL emergencies boot camp: using simulation to onboard residents. *Laryngoscope* 2011; **121**: 2114–21.
17 de Visser H, Watson MO, Salvado O, Passenger JD. Progress in virtual reality simulators for surgical training and certification. *Med. J. Am.* 2011; **194**: S38–40.
18 Kennedy CC, Maldonado F, Cook DA. Simulation-based bronchoscopy training: systematic review and meta-analysis. *Chest* 2013; **144**: 183–92.
19 Issenberg SB, McGaghie WC, Petrusa ER, Lee Gordon D, Scalese RJ. Features and uses of high-fidelity medical simulations that lead to effective learning: a BEME systematic review. *Med. Teach.* 2005; **27**: 10–28.
20 Davoudi M, Colt HG. Bronchoscopy simulation: a brief review. *Adv. Health Sci. Educ. Theory Pract.* 2009; **14**: 287–96.
21 Fielding D, Phillips M, Robinson P, Irving L, Garske L, Hopkins P, Thoracic Society of Australia and New Zealand. Advanced interventional pulmonology procedures: training guidelines from the Thoracic Society of Australia and New Zealand. *Respirology* 2012; **17**: 1176–89.
22 Wahidi MM, Silvestri GA, Coakley RD, Ferguson JS, Shepherd RW, Moses L, Conforti J, Que LG, Anstrom KJ, McGuire F *et al.* A prospective multicenter study of competency metrics and educational interventions in the learning of bronchoscopy among new pulmonary fellows. *Chest* 2010; **137**: 1040–9.
23 Bronchoscopy International. The bronchoscopy education project (part II): endobronchial ultrasound bronchoscopy competency program. 2011. [Accessed 25 Sep 2011.] Available from URL: http://www.aabronchology.org/PDF/bronch_course/BEP%20part%20II%20EBUS%20Training%20Manual.pdf
24 Unroe MA, Shofer SL, Wahidi MM. Training for endobronchial ultrasound: methods for proper training in new bronchoscopic techniques. *Curr. Opin. Pulm. Med.* 2010; **16**: 295–300.
25 Salamonsen M, McGrath D, Steiler G, Ware R, Colt H, Fielding D. A new instrument to assess physician skill at thoracic ultrasound, including pleural effusion markup. *Chest* 2013; **144**: 930–4.
26 Bronchoscopy International. Pleural ultrasound assessment tool. 2013. [Accessed 4 Feb 2014.] Available from URL: http://www.bronchoscopy.org/downloads/tools/USGSTAT%20assessment%20tool.pdf
27 Colt HG, Davoudi M, Quadrelli S, Rohani NZ. Use of competency-based metrics to determine effectiveness of a postgraduate thoracoscopy course. *Respiration* 2010; **80**: 553–9.
28 Ernst A, Silvestri GA, Johnstone D, American College of Chest Physicians. Interventional pulmonary procedures: guidelines from the American College of Chest Physicians. *Chest* 2003; **123**: 1693–717.
29 Wahidi MM, Silvestri G, Conforti J, Ferguson S, Coakley R, Patel C, Moses L, Downie G. Assessment of pulmonary fellows acquisition of bronchoscopy skills. *Chest* 2007; **132**: 514S–515S.
30 Downie G, Bowling M, Silvestri G, Ferguson S, Coakley R, Conforti J, Wahidi M. The Multi-State Bronchoscopy Education Project: acquisition of cognitive skills. *Chest* 2007; **132**: 664S.
31 Dalal PG, Dalal GB, Pott L, Bezinover D, Prozesky J, Bosseau Murray W. Learning curves of novice anesthesiology residents performing simulated fibreoptic upper airway endoscopy. *Can. J. Anesth.* 2011; **58**: 802–9.

32 Murin S, Stollenwerk NS. Simulation in procedural training: at the tipping point. *Chest* 2010; **137**: 1009–11.
33 Wahidi MM, Hulett C, Pastis N, Shepherd RW, Shofer SL, Mahmood K, Lee H, Malhotra R, Moser B, Silvestri GA. Learning experience of linear EBUS among pulmonary trainees. *Chest* 2014; **145**: 574–8.
34 Gladwell M. The 10 000 hour-rule. In: *Outliers*. Little, Bown and Co., New York, 2008; 35–68.
35 Iobst WF, Sherbino J, Cate OT, Richardson DL, Dath D, Swing SR, Harris P, Mungroo R, Holmboe ES, Frank JR. Competency-based medical education in postgraduate medical education. *Med. Teach.* 2010; **32**: 651–6.
36 Aggarwal R, Darzi A. Technical-skills training in the 21st century. *N. Engl. J. Med.* 2006; **335**: 2695–6.
37 Lund ME. Twenty-five questions: an important step on a critical journey. *Chest* 2009; **135**: 256–8.
38 Davoudi M, Wahidi MM, Rohani NZ, Colt HG. Comparative effectiveness of low- and high-fidelity bronchoscopy simulation for training in conventional transbronchial needle aspiration and user preferences. *Respiration* 2010; **80**: 327–34.
39 Colt HG, Davoudi M, Murgu S, Rohani NZ. Measuring learning gain during a one-day introductory bronchoscopy course. *Surg. Endosc.* 2011; **25**: 207–16.
40 Quadrelli S, Davoudi M, Galíndez F, Colt HG. Reliability of a 25-item low-stakes multiple-choice assessment of bronchoscopic knowledge. *Chest* 2009; **135**: 315–21.
41 Stratakos G. Contemporary bronchoscopy training and assessment: a la recherché du professionnalisme perdu? *Respiration* 2012; **83**: 101–2.
42 Nie XM, Cai G, Shen X, Yao XP, Zhao LJ, Huang Y, Han YP, Bai C, Li Q. Bronchoscopy in some tertiary grade A hospitals in China: two years' development. *Chin. Med. J.* 2012; **125**: 2115–19.
43 Davoudi M, Osann K, Colt HG. Validation of two instruments to assess technical bronchoscopic skill using virtual reality simulation. *Respiration* 2008; **76**: 92–101.
44 Hake RR. Interactive-engagement vs traditional methods: a six-thousand-student survey of mechanics test data for introductory physics courses. *Am. J. Phys.* 1998; **66**: 64–74.
45 Wagner M, Herth FJ, Stanzel F, Hautmann H, Eberhardt R, Becker HD, Hetzel J, Hetzel MR, Gesierich W, Ficker JH. [Recommendations for training courses in bronchoscopy]. *Pneumologie* 2011; **65**: 219–22.
46 Dai J, Shen YF, Sano Y, Li XB, Xue HB, Zhao YJ, Gao YJ, Song Y, Ge ZZ. Evaluation of narrow-band imaging in the diagnosis of colorectal lesions: is a learning curve involved? *Dig. Endosc.* 2013; **25**: 180–8.
47 Davoudi M, Colt HG, Osann KE, Lamb CR, Mullon JJ. Endobronchial ultrasound skills and tasks assessment tool. *Am. J. Respir. Crit. Care Med.* 2012; **186**: 773–9.
48 Zendejas B, Cook DA, Bingener J, Huebner M, Dunn WF, Sarr MG, Farley DR. Simulation-based mastery learning improves patient outcomes in laparoscopic inguinal hernia repair: a randomized controlled trial. *Ann. Surg.* 2011; **254**: 502–11.
49 Kane MT. The role of reliability in criterion-referenced tests. *J. Educ. Meas.* 1986; **23**: 221–4.
50 Wilcox RR. A note on the length and passing score of a mastery test. *J. Educ. Stat.* 1976; **1**: 359–64.
51 Jantz MA, McGaghie WC. It's time for a STAT assessment of bronchoscopy skills. *Am. J. Crit. Care Med.* 2012; **186**: 7004–705.
52 Konge L, Clementsen P, Larsen KR, Arendrup H, Buchwald C, Ringsted C. Establishing pass/fail criteria for bronchoscopy performance. *Respiration* 2012; **83**: 140–6.
53 Konge L, Clementsen P, Larsen KR, Arendrup H, Buchwald C, Ringsted C. Reliable and valid assessment of clinical bronchoscopy performance. *Respiration* 2012; **83**: 53–60.
54 Konge L, Arendrup H, Buchwald C, Ringsted C. Using performance in multiple simulated scenarios to assess bronchoscopy skills. *Respiration* 2011; **81**: 483–90.
55 McLaughlin JE, Roth MT, Glatt DM, Gharkholonarehe N, Davidson CA, Griffin LM, Esserman DA, Mumper RJ. The flipped classroom: a course redesign to foster learning and engagement in a health professions school. *Acad. Med.* 2014; **89**: 236–43.
56 Periyakoil VS, Basaviah P. The flipped classroom paradigm for teaching palliative care skills. *Virtual Mentor* 2013; **15**: 1034–7.
57 Kruglikova I, Grantcharov TP, Drewes AM, Funch-Jensen P. The impact of constructive feedback on training in gastrointestinal endoscopy using high-fidelity Virtual-Reality simulation: a randomised controlled trial. *Gut* 2010; **59**: 181–5.
58 Mehta NB, Hull AL, Young JB, Stoller JK. Just imagine: new paradigms for medical education. *Acad. Med.* 2013; **88**: 1418–23.
59 American College of Chest Physicians. Lung cancer education courses. 2013. [Accessed 2 Jan 2014.] Available from URL: http://www.chestnet.org/Education/Products/Live-Learning
60 Houston M, Lin L. Humanizing the classroom by flipping the homework versus lecture equation. In: Resta P (ed) *Proceedings of Society for Information Technology & Teacher Education International Conference 2012*. AACE, Chesapeake, VA, 2012; pp. 1177–82. [Accessed 8 Feb 2014.] Available from URL: http://www.editlib.org/p/39738
61 Savery JR, Duffy TM Problem Based Learning: an instructional model and its constructivist framework. CRLT Technical Report no. 16-01; June, 2001.
62 Jonsen AR, Siegler M, Winslade WJ. *Clinical Ethics*, 6th edn. McGraw Hill, New York, 2006.
63 Colt H, Murgu S. *Bronchoscopy and Central Airway Disorders: A Patient-Centered Approach: Expert Consult Online and Print*, 1st edn. Elsevier, Philadelphia, PA, 2012.
64 Bronchoscopy International. What is the practical approach to interventional bronchoscopy: bronchoscopy international. 2013. [Accessed 1 Feb 2014.] Available from URL: http://www.bronchoscopy.org/practical-approach
65 ACGME. ACGME competencies. 2013. [Accessed 1 Feb 2014.] Available from URL: http://www.acgme.org
66 Miller R, Murgu S, Colt H Practical approach to bronchoscopy. 2012. [Accessed 1 Feb 2014.] Available from URL: http://www.youtube.com/watch?v=CNi_CBFKpiM
67 Faculty development program and introduction to flexible bronchoscopy. WABIP project in Tokyo, Japan 15–17 Mar 2013.
68 Faculty development program: a Bronchoscopy International project, Mayo Clinic, Rochester, MN, 17–18 Aug 2012.
69 Mercier EM, Higgins SE. Collaborative learning with multi-touch technology: developing adaptive expertise. *Learn. Instr.* 2013; **25**: 13–23.
70 Kohn L, Corrigan JM, Donaldson MS (eds). *To Err is Human: Building a Safer Health System*, Institute of Medicine. National Academy Press, Washington, DC, 1999.
71 Deutsch ES, Christenson T, Curry J, Hossain J, Zur K, Jacobs I. Multimodality education for airway endoscopy skill development. *Ann. Otol. Rhinol. Laryngol.* 2009; **118**: 81–6.
72 Stather DR, MacEachern P, Chee A, Dumoulin E, Tremblay A. Evaluation of clinical endobronchial ultrasound skills following clinical versus simulation training. *Respirology* 2012; **17**: 291–9.
73 Stather DR, Maceachern P, Rimmer K, Hergott CA, Tremblay A. Assessment and learning curve evaluation of endobronchial ultrasound skills following simulation and clinical training. *Respirology* 2011; **16**: 698–704.
74 Marsland CP, Robinson BJ, Chitty CH, Guy BJ. Acquisition and maintenance of endoscopic skills: developing an endoscopic dexterity training system for anesthesiologists. *Clin. Anesth.* 2002; **14**: 615–19.
75 Agrò F, Sena F, Lobo E, Scarlata S, Dardes N, Barzoi G. The Dexter Endoscopic Dexterity Trainer improves fibreoptic bronchoscopy skills: preliminary observations. *Can. J. Anaesth.* 2005; **52**: 215–16.
76 Hilmi OJ, White PS, McGurty DW, Oluwole M. Bronchoscopy training: is simulated surgery effective? *Clin. Otolaryngol. Allied Sci.* 2002; **27**: 267–9.

77 Blum MG, Powers TW, Sundaresan S. Bronchoscopy simulator effectively prepares junior residents to competently perform basic clinical bronchoscopy. *Ann. Thorac. Surg.* 2004; **78**: 287–91.
78 Ameur S, Carlander K, Grundström K, Hallberg P, Lundgren K, Lundquist PG, Wikström T. [Learning bronchoscopy in simulator improved dexterity rather than judgement]. *Lakartidningen* 2003; **100**: 2694–9.
79 Fann JI, Feins RH, Hicks GL Jr, Nesbitt JC, Hammon JW, Crawford FA Jr. Senior tour in cardiothoracic surgery. Evaluation of simulation training in cardiothoracic surgery: the Senior Tour perspective. *J. Thorac. Cardiovasc. Surg.* 2012; **143**: 264–72.
80 Bronchoscopy International. *Faculty Development Program Training Manual*. Bronchoscopy International, Laguna Beach, CA, 2012.

